IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-106

1. Topic (12 words max)

AI-Based Perinatal Depression Screening and Support System

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-106
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Postpartum Depression (PPD) is a prevalent and serious mental health condition affecting mothers during the postnatal period, with global prevalence estimates ranging from 10% to 20% [1], [2]. PPD negatively impacts maternal emotional well-being, infant development, family relationships, and long-term social stability. Despite its severity, PPD often remains underdiagnosed due to stigma, lack of awareness, limited access to mental health professionals, and high costs associated with clinical consultations—particularly in low- and middle-income countries such as Sri Lanka [3], [4].

The primary group affected by this problem includes postpartum mothers, especially first-time mothers and those with limited social or emotional support. Untreated PPD can lead to chronic depression, impaired mother–child bonding, reduced quality of childcare, and increased healthcare expenditure, creating significant social and economic burdens [5]. From an ethical and legal perspective, inadequate access to early mental health screening and supportive care may violate a mother's right to timely and equitable healthcare services [6].

Although mobile health applications for mental health have grown rapidly, most existing systems focus on general depression, anxiety, or stress and fail to address postpartum-specific conditions such as recovery stage, physical readiness, emotional vulnerability, and daily routine challenges [7], [8]. Furthermore, many applications lack personalized, adaptive support and rely heavily on self-initiated help-seeking, which is often unrealistic for emotionally distressed mothers [9]. Research indicates that AI-based screening and recommendation systems can significantly improve early detection, personalization, and accessibility of mental health support when designed with ethical and safety constraints [10], [11].

This research aligns with the AI and Data Science research cluster and targets the Digital Health and Healthcare Technology industry vertical. By supporting early screening, emotional well-being, safe postpartum self-care, and routine adherence, the proposed system contributes directly to UN Sustainable Development Goal 3 (Good Health and Well-Being) and SDG 5 (Gender Equality) by promoting maternal mental health and empowering women through accessible digital healthcare solutions [12].

References

[1] World Health Organization, *Maternal mental health*, WHO, 2023.

[2] O'Hara, M. W., & McCabe, J. E., "Postpartum depression: Current status and future directions," *Annual Review of Clinical Psychology*, 2013.

[3] World Health Organization, *Mental Health Atlas*, 2022.

[4] Stewart, D. E., et al., "Postpartum depression: Literature review of risk factors and interventions," *Harvard Review of Psychiatry*, 2003.

[5] Murray, L., & Cooper, P. J., "Effects of postnatal depression on infant development," *Psychological Medicine*, 1997.

[6] United Nations, *Universal Declaration of Human Rights*, Article 25, 1948.

[7] Chandrashekar, P., "Do mental health mobile apps work? Evidence and recommendations," *mHealth*, 2018.

[8] E. S. Mendis et al., "Mobile Application for Mental Health Using Machine Learning," *IEEE International Conference on Advancements in Computing (ICAC)*, 2022.

[9] Torous, J., et al., "Digital mental health and engagement challenges," *World Psychiatry*, 2020.

[10] Shatte, A. B. R., et al., "Machine learning in mental health," *JMIR Mental Health*, 2019.

[11] Topol, E., *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*, Basic Books, 2019.

[12] United Nations, *Sustainable Development Goals*, 2015.

6. Brief description of Existing Research and Systems (300 words max)

Current digital solutions for postpartum depression mainly focus on *awareness and basic screening*. Mobile applications and websites such as BabyCenter, What to Expect, and general wellness platforms like Headspace or Calm provide educational articles, mood check-ins, and relaxation content. Some maternal health apps integrate EPDS-style questionnaires, but these are usually conducted only once or at long intervals. The results are often static and do not guide users beyond simple advice such as "seek help if symptoms persist." These systems do not continuously monitor mothers, do not analyze long-term patterns, and rarely provide personalized, postpartum-specific guidance. Most importantly, they operate in isolation from healthcare providers and lack any form of predictive intelligence.

Research in this field shows that early detection using EPDS/PHQ-9 is effective, and recent studies demonstrate the value of machine learning for depression classification, NLP for analyzing diary text, and time-series models for predicting symptom trends. Separate research streams have also explored chatbots and digital interventions for emotional support. However, these efforts remain fragmented: one system may detect risk, another may provide content, and another may experiment with chatbots. Very few systems combine *screening, prediction, personalization, and interaction* into a single continuous care loop.

This fragmentation creates significant gaps. Existing platforms:

- Provide only snapshot-based assessments, not continuous monitoring
- Lack future risk prediction
- Offer generic, non-adaptive content
- Do not tailor guidance to postpartum-specific needs
- Do not generate clinical reports for doctors
- Fail to act as a daily companion for mothers

Our proposed system directly addresses these limitations. It unifies four intelligent components—screening, trend analysis with prediction, personalized emotional guidance, and an AI chatbot—into one coherent platform. Unlike existing apps, it continuously learns from user inputs, predicts future risk, adapts recommendations dynamically, and bridges the gap between mothers and clinicians through structured reports. This transforms postpartum care from reactive and fragmented into proactive, personalized, and supportive. The system is not merely an information source; it becomes a smart, empathetic companion that empowers mothers, supports early intervention, and enhances clinical decision-making.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

This research proposes an **AI-powered Postpartum Mental Health Support System** designed to assist new mothers during the critical postpartum period through **early detection, personalized guidance, and continuous non-clinical support**. The solution integrates machine learning, recommendation systems, and conversational AI to provide a holistic, ethical, and scalable digital companion for postpartum well-being.

The system begins with **structured daily data collection** from postpartum mothers, including standardized psychological questionnaire responses (EPDS/PHQ-9), self-reported symptoms, lifestyle indicators, preferences, and routine-related inputs. This data forms the foundation for all subsequent intelligent processing.

Conceptual Architecture Overview

As illustrated in the conceptual diagram, the system is composed of **four tightly integrated AI-driven components**, each handled by a dedicated research member:

1. **Depression Screening & Classification Module**
   This module analyzes validated psychological assessment data and symptom patterns using supervised machine learning models such as Logistic Regression, Random Forest, or SVM. It classifies mothers into **Low, Medium, or High postpartum depression risk levels**, enabling early awareness and timely escalation when necessary.

2. **Gentle Emotional Support Recommendation System**
   For mothers classified as **Low or Medium risk**, this component provides **safe, non-medical emotional support**. Using an expert-driven dataset and a trained recommendation model, it personalizes calming content such as Buddhist bana, motivational videos, mandala coloring, breathing audio, and relaxing activities based on risk level and individual preferences. The output is a **daily emotional support plan** that promotes calmness, hope, and emotional stability.

3. **AI-Guided Postpartum Exercise Recommendation System**
   This component focuses on **safe, non-clinical postpartum exercises** suitable only for Low and Medium risk mothers. Using structured postpartum health indicators (e.g., weeks after delivery, energy level, pain indicators, delivery type), an explainable ML model recommends gentle activities such as breathing exercises or light stretching. All recommendations are constrained by **clinically informed safety rules**, ensuring no medical treatment or physical therapy is provided.

4. AI Chatbot & Routine Management Module

Acting as a continuous companion, the chatbot enables mothers to ask questions related to postpartum well-being, receive emotional guidance, and manage daily routines. It generates personalized schedules and sends timely reminders (e.g., meals, rest, exercises), reinforcing healthy habits and consistency.

Outcome

Together, these components produce an integrated system that delivers:

- Depression risk level with confidence scoring
- Personalized emotional and exercise support plans
- Symptom trend tracking and visualization
- AI chatbot assistance and routine reminders
- Doctor-ready mental health summaries and alerts

Overall, the solution bridges **early detection and actionable daily support**, extending postpartum care beyond clinical settings while maintaining strict non-medical and ethical boundaries.

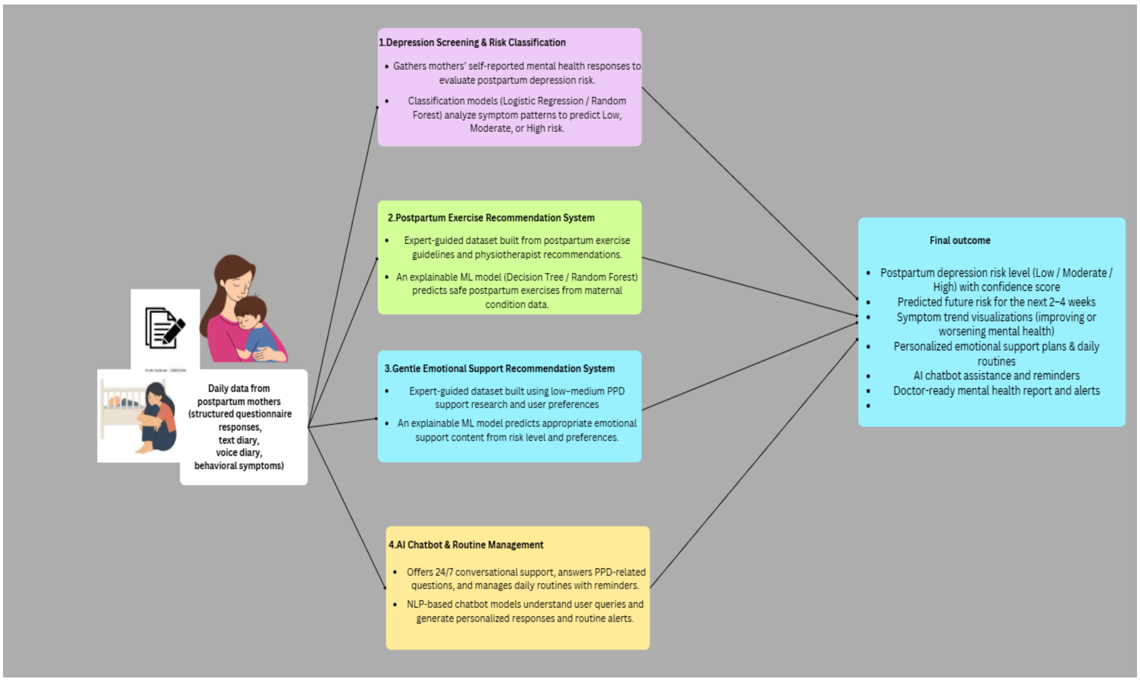

Figure 1: conceptual diagram

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Postpartum depression (PPD) is a widely researched maternal mental health condition with strong multidisciplinary knowledge support from psychiatry, psychology, obstetrics, physiotherapy, public health, and artificial intelligence research. The proposed system benefits from the availability of validated clinical guidelines, standardized screening tools, expert-driven recovery frameworks, and established AI and machine learning techniques, ensuring the feasibility and academic grounding of the research.

Clinical and Psychological Domain Expertise

Extensive domain expertise is available in maternal mental health through psychiatrists, clinical psychologists, obstetricians, and midwives who routinely manage postpartum depression. International guidelines published by organizations such as the World Health Organization (WHO), DSM-5, and national maternal health authorities clearly define postpartum depression symptoms, severity levels, and risk thresholds. This expertise supports the depression screening and risk classification component, enabling accurate categorization of mothers into Low, Medium, or High risk levels while ensuring ethical and clinically safe system behavior.

Standardized Screening Knowledge and Assessment Instruments

Well-validated screening tools such as the Edinburgh Postnatal Depression Scale (EPDS) and PHQ-9 are widely used in both clinical practice and research. These instruments provide structured, labeled psychological assessment data suitable for supervised machine learning models. Their reliability, sensitivity, and specificity are well documented, allowing effective feature engineering, model training, and evaluation for postpartum depression risk classification.

Expert Knowledge and Data for Safe Postpartum Exercise Recommendation

Postpartum physical recovery and safe movement are supported by extensive research and physiotherapy guidelines. Clinical literature and expert recommendations clearly identify low-risk, non-medical exercises suitable for postpartum mothers at different recovery stages. This includes breathing exercises, gentle stretching, and relaxation movements for mothers classified as Low or Medium risk. Expert-driven rules and postpartum recovery frameworks allow the creation of structured, explainable datasets for training machine learning models to recommend safe exercise types without diagnosing conditions or providing medical treatment. The availability of physiotherapist-approved recovery guidelines ensures safety and ethical compliance.

Knowledge Base for Emotional Support and Non-Medical Interventions

Evidence-based emotional support methods for postpartum mothers—such as guided breathing, mindfulness, motivational content, calming audio, and low-cognitive-load activities—are well documented in psychological research. Expert-validated emotional support techniques provide a reliable foundation for building structured content libraries and training recommendation models that align with individual preferences and mental health risk levels. This supports personalized, non-clinical emotional well-being interventions for Low and Medium risk mothers.

AI, Chatbot, and Routine Management Expertise

There is strong technical maturity in conversational AI, recommendation systems, habit formation, and routine management. Natural language processing, rule-based scheduling, explainable machine learning models (e.g., Decision Trees and Random Forests), and notification systems are well-established technologies. These enable the development of a postpartum-specific conversational assistant that provides accurate information, personalized daily routines, and real-time reminders based on user context and risk classification.

Data Governance and Ethical Considerations

Well-established ethical frameworks exist for handling sensitive maternal mental health data, including informed consent, data anonymization, and privacy protection. These guidelines ensure responsible data usage, secure system design, and clinical acceptability.

Overall, the availability of multidisciplinary clinical expertise, validated assessment tools, expert-driven recovery knowledge, accessible datasets, and mature AI techniques provides a strong and realistic foundation for developing the proposed AI-based Postpartum Depression Support System

9. Objectives and Novelty

Main Objective To design and develop an AI-based postpartum depression support system that screens and classifies depression risk levels, and provides personalized emotional support, safe postpartum exercise recommendations, and AI-assisted daily routine guidance to support maternal mental well-being in a non-clinical and ethically safe manner.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
K H P T Fernando IT22119476	To design and implement an AI-based postpartum depression screening and risk classification module that combines standardized questionnaire responses with behavioral indicators to classify postpartum mothers into Low, Medium, or High risk categories, while incorporating daily	• Identify and analyze postpartum behavioral indicators such as sleep, mood, stress, appetite, and energy levels to assess depression risk. • Design structured input forms for screening and daily diary entries, ensuring diary data supports trends but does not directly determine risk. • Implement a daily diary feature for mothers to record non-clinical emotional and lifestyle information to monitor patterns over time.	• Introduces a multi-factor AI-based postpartum depression screening approach that goes beyond fixed thresholds by learning patterns across behavioral and emotional indicators. • Combines structured screening inputs with daily diary and voice-to-text entries to support continuous monitoring without misusing diary data for direct risk decisions.

	diary entries and voice-to-text inputs for continuous monitoring and ethical decision support without providing medical diagnosis.	• Integrate voice recognition to convert user-recorded voice notes into text, simplifying diary maintenance. • Preprocess collected data by normalizing values and encoding features for machine learning analysis. • Develop and train a supervised machine learning model to classify postpartum depression risk into Low, Medium, or High categories. • Implement a user-friendly system with backend APIs and frontend interfaces, showing risk results, diary summaries.	• Provides probabilistic risk classification (Low, Medium, High) instead of rigid score-based decisions, handling ambiguity in early or borderline postpartum depression cases. • Incorporates voice-enabled diary recording to improve accessibility and reduce user burden, particularly for mothers experiencing fatigue or stress. • Designed as a non-diagnostic, decision-support tool that empowers users to self-monitor and reflect on their emotional health.
De Silva S.R.N IT22284884	Develop a Gentle Emotional Support Recommendation System for *Low and Medium* postpartum depression mothers that provides safe, non-medical emotional guidance.	• **Collect expert knowledge:** Interview doctors, counselors, and review research to identify emotional support methods safe for Low & Medium PPD mothers • **Content library creation:** Gather and organize videos, audio, mandala art, guided	• Provides **personalized emotional support** based on both risk level and user preferences, unlike generic apps that give the same content to all mothers. • **Custom-built dataset and ML model:** Instead of using generic or public datasets,

		breathing exercises, motivational clips, and games. Verify sources for credibility and appropriateness. • **Preference-based input system:** Design structured input forms where mothers indicate their preferences (likes/dislikes for bana, music, art, meditation, games). • **Dataset creation:** Build a structured dataset mapping *risk level + preferences → most suitable emotional content*, with 200–300 entries to cover various combinations. • **ML model training:** Train a Decision Tree or Random Forest model in Python (scikit-learn) using the dataset to predict the best content for each mother. • **Backend implementation:** Develop API logic to receive user input, predict the recommended content, and generate a daily emotional support plan.	the system uses expert-verified rules to train a model specifically for postpartum emotional support. • **Dynamic daily emotional plan:** Generates a daily routine for emotional support, with scheduled content recommendations, unlike most systems that only suggest content statically. • **Safe, non-medical, culturally sensitive content:** Includes Buddhist bana, motivation, mandala coloring, calming games—carefully selected to avoid physical strain or medical intervention, which is rarely addressed in existing systems. • Bridges **screening/risk classification with actionable emotional support**, creating a seamless path from detection to daily well-being. • Incorporates AI/ML to **adapt recommendations** over time, enhancing personalization

		• . **Frontend integration:** Display the daily plan, links to videos, audio, and activities in an easy-to-use interface with clear timing and suggestions. • **Feedback mechanism:** Include a simple “👍 / 😐 / 👎 ” feedback option so AI can adjust content suggestions in future iterations (optional adaptive learning).	beyond static rule-based systems. • Supports **low-resource and home-based mental health care**, providing accessible, 24/7 support to new mothers.
Perera K.T.S.N IT22187628	To design and implement an AI-guided postpartum exercise recommendation system that provides safe, personalized, and adaptive exercise guidance for low and medium risk postpartum mothers by learning from structured postpartum health indicators and clinically informed training data, while ensuring all recommendations	• Collect and analyze postpartum exercise safety guidelines from doctors, physiotherapists, and trusted medical literature to identify safe exercise categories, intensity limits, and recovery timelines for postpartum mothers. • Identify and define structured postpartum health indicators such as weeks after delivery, energy level, presence of back or pelvic pain, delivery type, and physical readiness that influence exercise suitability. • Design and construct an expert-driven training dataset that maps postpartum conditions	• Introduces an AI-guided postpartum exercise recommendation system that focuses specifically on postpartum recovery conditions rather than general fitness or wellness, addressing a clinically sensitive and under-researched domain. • Utilizes an expert-constructed training dataset derived from medical knowledge and postpartum recovery logic, enabling machine learning in a domain where real-world labeled datasets are scarce or unavailable.

	remain within medically safe and non-clinical boundaries.	and recovery stages to the safest recommended exercise types, based on clinical reasoning and postpartum recovery logic. • Develop a machine learning model using explainable algorithms to learn the relationship between postpartum conditions and safe exercise recommendations from the constructed dataset. • Implement a safety-aware decision pipeline where machine learning predictions are constrained by clinically defined rules to ensure that high-risk conditions, such as early post-delivery stages or C-section recovery, override all other recommendations. • Design and integrate a curated library of doctor-approved postpartum exercise and breathing videos from free and reliable sources, annotated with metadata including exercise type, intensity,	• Combines machine learning–based prediction with clinically inspired safety constraints, ensuring adaptability while preventing unsafe or inappropriate exercise recommendations. • Employs explainable AI techniques to support transparency and trust in exercise recommendations, which is critical for healthcare-related applications. • Provides adaptive personalization through controlled feedback mechanisms that adjust recommendations without violating medical safety boundaries. • Integrates structured multimedia exercise guidance using metadata-driven selection, ensuring that only safe, relevant, and professionally aligned content is delivered to users.

		duration, and applicable conditions. • Develop an adaptive personalization mechanism that modifies exercise duration and selection based on user feedback while strictly maintaining predefined safety limits. • Implement backend services to handle data processing, AI prediction, and safety validation, and frontend components to present daily exercise plans, guidance videos, and progress information in a clear and supportive manner. • Integrate the machine learning model, safety rules, content library, and user interface into a unified system and conduct scenario-based testing to evaluate recommendation accuracy, safety compliance, and system reliability.	• Extends postpartum care beyond clinical settings by offering continuous, at-home digital support that complements professional guidance without replacing medical expertise.
Jayawardhana W.M.K.D IT22186324	To develop an AI-powered conversational assistant with daily	• Design and implement an AI-powered conversational assistant focused on	• Introduces a postpartum depression–specific AI chatbot grounded in maternal mental-health

	routine planning that supports mothers experiencing postpartum depression by providing mental health guidance, personalized routines, timely reminders, and information on common newborn diseases, early-life health conditions, and basic care practices during the first few months after birth, using both text and voice-based user interactions.	postpartum depression support. • Enable natural language interaction to answer questions related to emotional well-being, self-care, and daily lifestyle management. • Develop a personalized daily routine planning feature based on user preferences and mental health status. • Implement a smart reminder and notification system to support routine adherence and self-care activities. • Adapt chatbot responses and routine recommendations dynamically using user risk levels and symptom trends.	knowledge and clinical guidelines, supporting both text and voice-based interactions to enable natural communication for postpartum mothers. • Employs a RAG-based response mechanism to generate trustworthy, medically grounded answers by retrieving validated postpartum depression information, reducing AI hallucinations and ensuring healthcare safety. • Integrates conversational AI with personalized daily routine planning, transforming mental-health guidance into actionable schedules that include self-care, rest, exercise, and therapeutic activities. • Adapts chatbot responses and routine recommendations dynamically using multimodal user inputs (text and voice), individual risk levels, mood

			trends, and symptom changes to provide context-aware, empathetic support. • Implements smart reminders and notifications to reinforce routine adherence, emotional regulation, and daily consistency, delivering continuous personalized support between clinical consultations and enhancing user engagement

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Instructed the students to collect real time data.

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Sanvitha	Kasthurirarachchi	[illegible]
Co-Supervisor	Ms.	Sanjeevi	Chandrasiri	Sanjeevi
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				